Yearim Kim

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Research Summary

I build **mechanistic interpretability** methods that explain *how* vision models compute, not merely *where* they look. My work introduces the **instance-specific effective receptive field (iERF)** as a unifying primitive that connects local attribution, global concept structure, and circuit-level analysis in a single framework—first presented at **ICLR (2024)** and extended into **IEEE TPAMI (2026)**. Beyond post-hoc explanation, I design **interpretable-by-design** architectures: **Bi-ICE (WACV 2026)** grounds predictions in human-understandable concepts through bidirectional interactions between concept and input embeddings, making each concept's contribution traceable inside the model, and **HIVE** (under review) extends this direction to compositional structure by inducing hierarchical visual embeddings that capture how parts form wholes. Both lines were developed through an international collaboration with **Arizona State University**. Broadly, I aim to make large vision and multimodal models *auditable*: faithful explanation, concept discovery, and machine unlearning.

Keywords: Mechanistic Interpretability • Explainable AI • Vision Transformers • Sparse Autoencoders • Concept Learning • Machine Unlearning

Education

Seoul National University — Seoul, Republic of Korea
Integrated M.S./Ph.D. in Artificial Intelligence — *Sep 2021 – Present*

- Advisor: Prof. **Nojun Kwak**, Machine Intelligence & Pattern Analysis Lab (MIPAL).
- Thesis: *Mechanistic Interpretability of Vision Models via Effective Receptive Fields.*

Korea University — Seoul, Republic of Korea
B.S. in Statistics — *Mar 2017 – Feb 2021*

Selected Publications

▶ **TPAMI 2026** **From Local to Global to Mechanistic: An iERF-Centered Unified Framework for Interpreting Vision Models**

Yearim Kim*, Sangyu Han*, Nojun Kwak. *IEEE Trans. Pattern Analysis and Machine Intelligence*, 2026 (to appear).

Primary work. Unifies local attribution, global concept analysis, and mechanistic circuit tracing under a single effective-receptive-field formulation—the journal-length extension of the ICLR 2024 line of work below.

▶ **ICLR 2024** **Respect the Model: Fine-grained and Robust Explanation with Sharing Ratio Decomposition**

Sangyu Han*, **Yearim Kim***, Nojun Kwak. *International Conference on Learning Representations*, 2024.

Model-faithful attribution that is robust to adversarial explanation attacks; foundation of the iERF framework.

▶ **WACV 2026** **Bi-ICE: An Inner Interpretable Framework for Image Classification via Bi-directional Interactions between Concept and Input Embeddings**

Jinyung Hong*, **Yearim Kim***, Keun Hee Park, Sangyu Han, Nojun Kwak, Theodore P. Pavlic. *IEEE/CVF Winter Conf. on Applications of Computer Vision*, 2026.

Interpretable-by-design concept module.

Publications

* equal contribution. 14 papers total: 2 journal, 8 conference/workshop, 4 under review.

Journal Articles

[J2] **Yearim Kim***, Sangyu Han*, Nojun Kwak. "From Local to Global to Mechanistic: An iERF-Centered Unified Framework for Interpreting Vision Models." **IEEE TPAMI**, 2026. *(to appear)*

[J1] Jaeseok Choi, Yeji Song, **Yearim Kim**. "MD3D: Mixture-Density-Based 3D Object Detection in Point Clouds." **IEEE Access**, 2022.

Conference & Workshop Papers

[C9] **Yearim Kim**, Jinyung Hong, Nojun Kwak. "Separately Capable, Jointly Failing: Recognition Without Functional Factual Access in Vision–Language Models ." **Latent Visual Reasoning (LVR) Workshop @ WACV**, 2027.

[C8] Daewon Yoon, Injun Baek, Sangyu Han, **Yearim Kim**, Nojun Kwak. "VDPP: Video Depth Post-Processing for Speed and Scalability." **ECV Workshop @ CVPR**, 2026.

[C7] **Yearim Kim***, Injun Baek*, Nojun Kwak. "When Saying No Makes Better Videos: Designing Dual Gatekeeping for Pedagogically Grounded AI Content Creation." **CHI Workshop**, 2026.

[C6] Injun Baek*, **Yearim Kim***, Nojun Kwak. "PedaCo-Gen: Scaffolding Pedagogical Agency in Human–AI Collaborative Video Authoring." **CHI EA**, 2026.

[C5] Jinyung Hong*, **Yearim Kim***, Keun Hee Park, Sangyu Han, Nojun Kwak, Theodore P. Pavlic. "Bi-ICE: An Inner Interpretable Framework for Image Classification via Bi-directional Interactions between Concept and Input Embeddings." **WACV**, 2026.

[C4] Sangyu Han, **Yearim Kim**, Nojun Kwak. "Causal Interpretation of Sparse Autoencoder Features in Vision." **eXCV Workshop @ ICCV**, 2025.

[C3] Sangyu Han*, **Yearim Kim***, Nojun Kwak. "Respect the Model: Fine-grained and Robust Explanation with Sharing Ratio Decomposition." **ICLR**, 2024.

[C2] Junhoo Lee, **Yearim Kim**, Nojun Kwak. "Any-Way Meta Learning." **AAAI**, 2024.

[C1] Yeonjin Chang, **Yearim Kim**, Nojun Kwak. "Fast Sun-Aligned Outdoor Scene Relighting Based on TensoRF." **WACV**, 2024.

Preprints & Under Review

[P4] Injun Baek*, Hyeongseok Lee*, **Yearim Kim**, Junhoo Lee, Nojun Kwak. "LandingAgent: A Reference-Annotated Dataset and Agentic Generation Framework for Landing Pages." *Under review.*

[P3] Gyomin Hwang*, Hyeonjin Kim*, Hyeonho Lee*, **Yearim Kim***, Yeji Song, Nojun Kwak. "μ^2-Bench: A Multilingual Machine Unlearning Benchmark." *Under review.*

[P2] Jinyung Hong*, **Yearim Kim***, Keun Hee Park, Pavan K. Turaga, Nojun Kwak, Andrew I. Yang. "HIVE: Hierarchical Induction of Visual Embeddings for Part–Whole Representation Learning." *Under review.*

[P1] Sangyu Han*, **Yearim Kim***, Nojun Kwak. "Beyond Activation Patches: ERF-Grounded Autolabeling of Sparse Autoencoder Features in Vision Transformers." *Under review.*

Research Experience

Graduate Researcher — MIPAL, Seoul National University Seoul, Korea
Advisor: Prof. Nojun Kwak *Jan 2021 – Present*

- Student Lab Lead (2024 – 2025): Coordinated a lab of about 40 members, managed shared GPU resources, and oversaw lab operations and project coordination.

- Lead the lab's interpretability research line; **first/co-first author on 8 papers** including IEEE TPAMI, ICLR, and WACV.
- Established international collaborations with **Arizona State University** (Bi-ICE, HIVE) spanning concept learning and part–whole representation.

Data Science Intern — Superbin Inc. Seongnam, Korea
Supervisor: Jungeun Jeong *Jul 2020 – Dec 2020*

- Analyzed deep-learning classification models for recyclable-material recognition deployed in circular-resource collection systems.

Projects & Grants

TexCorps 2026 — Ministry of Science and ICT (MSIT), Korea
Research-to-market translation program; XAI-based diagnostic agent *Jan 2026 – Aug 2026*

- Led market validation and customer discovery to commercialize laboratory interpretability research into a diagnostic agent tool.

Sketcher X — Exhibited at CES 2023, Las Vegas
Humanoid robotic portrait-drawing system *Nov 2022 – Jan 2023*

- Built the end-to-end framework enabling a robotic arm to sketch a person's portrait while holding a real-time conversation; demonstrated at **CES 2023**.

Teaching

Teaching Assistant — *Interpretability for Artificial Intelligence*
Samsung AI Expert Program *Sep 2024, Sep 2025, Jun 2026*

Teaching Assistant — *Linear Algebra for Artificial Intelligence*
Hyundai Heavy Industries *Nov–Dec 2022, Jun–Jul 2024*

Technical Skills & Languages

Programming Python, R
Languages Korean (native), English (advanced), Japanese (intermediate)